Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Sorensen, Christopher Ph.D.		
Project Title:	Aggregation in Dense Systems: Crossover from Cluster-Cluster-Aggregation in Percolation		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	66506-2601	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Chakrabarti, Amit (Kansas State University)		
Grant/Contract No.:	NNC04GA74G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are studying nanometer sized aerosols that aggregate and thus become so dense that they eventually form a gel. Gelation of an aerosol is very novel and we discovered this phenomenon. We will study the kinetics of dense aerosol aggregation and gelation. we will do this with both experiments and computer simulations and compare these techniques. We will stuy the properties of the aerosol gels formed and develop their potential as useful materials. Aerosols are a very common "phase" of matter, from aeosols in our atmosphere that effect our global cimate to other planetary almospheres, such as the sand storms of Mars or the thick atmosphere of Saturn's moon Titan. All aerosols are inherently unstable and hence aggregate with time. Thus it is important to understand the mechanisms of aggregation and the structures that result. This understnding, which we are pursuing with our research, will help us deal with various planetary atmospheres as we explore the solar system. A second source of significance of our work is the unexpected		

	discovery in our lab of novel ultralow density materials formed during aerosol gelation. These materials, which we call" aerosol gels", are similar to well-known aerogels but easier to make. Hence with further research may prove to have great technological utility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As described above, our research lays a fundamental understanding of aerosols which affect our global climate and visibility. Our discovery of aerosol gels may have great technological utility.
Task Progress:	We have used gelation in the aerosol phase to make what we call "aerososl gels". These are very low in density as low as 2.5mg/cc. We think they may have important technical applications similar to aerogel materials. We have also seen in both experiment and simulations what we call " superaggregates, which are aggregates of one fractal dimension composed of smaller aggregates of a different fractal dimension. this is the dominant morphology of gelation. We are also making progress on understanding the patterns in the functional behavior of light scattering from particles.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Other Journals or Periodicals	Fry D, Pierce F, Sorensen CM, Chakrabarti A. "Kinetics of aggregation in dense particulate systems" Physical Review E, submitted , Jan-2004
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Articles in Peer-reviewed Journals	Sorensen CM, Kim W, Fry D, Chakrabarti A. "Observation of soot superaggregates with a fractal dimension of 2.6 in laminar acetylene/air diffusion flame" Langmuir, 2003 Jul;19(18):7560–3. http://dx.doi.org/10.1021/la034063h , Jul-2003
Articles in Peer-reviewed Journals	Fry D, Chakrabarti A, Kim W, Sorensen CM "Structural crossover in dense irreversibly aggregating particulate system." Phys. Rev. E 2004 Jun;69(6):061401. https://doi.org/10.1103/PhysRevE.69.061401 ; PMID: 15244562 , Jun-2004
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Articles in Peer-reviewed Journals	Kim W, Sorensen CM, Chakrabarti A. "Universal occurrence of soot aggregates with a fractal dimension 2.6 in heavily sooting laminar diffusion flames." Langmuir. 2004 May 11;20(10):3969-73. http://dx.doi.org/10.1021/la036085+ ; PMID: 15969387 , May-2004
Articles in Peer-reviewed Journals	Fry D, Mohammad A, Chakrabarti A, Sorensen CM. "Cluster shape anisotropy in irreversibly aggregating particulate systems." Langmuir. 2004 Aug 31;20(18):7871-9. http://dx.doi.org/10.1021/la0494369 ; PMID: 15323542. (Originally reported in 2005 as "in press, January 2004.") , Aug-2004
Articles in Peer-reviewed Journals	Cerda JJ, Sintes T, Sorensen CM, Chakrabarti A. "Structure factor scaling in colloidal phase separation" Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Nov;70(5 Pt 1):051405. http://dx.doi.org/10.1103/PhysRevE.70.051405 ; PMID: 15600616. (Originally reported in 2005 as "in press January 2004.") , Nov-2004
Articles in Peer-reviewed Journals	Kim,W, Sorensen CM, Fry D, Chakrabarti A. "Soot aggregates, superaggregates and gel-like networks in laminar diffuison flames" Journal of Aerosol Science, submitted , Jan-2004